

One autonomous SDLC. One evidence-bound state machine.

Signal → knowledge → plan → Fleet execution → independent proof → release → runtime learning.

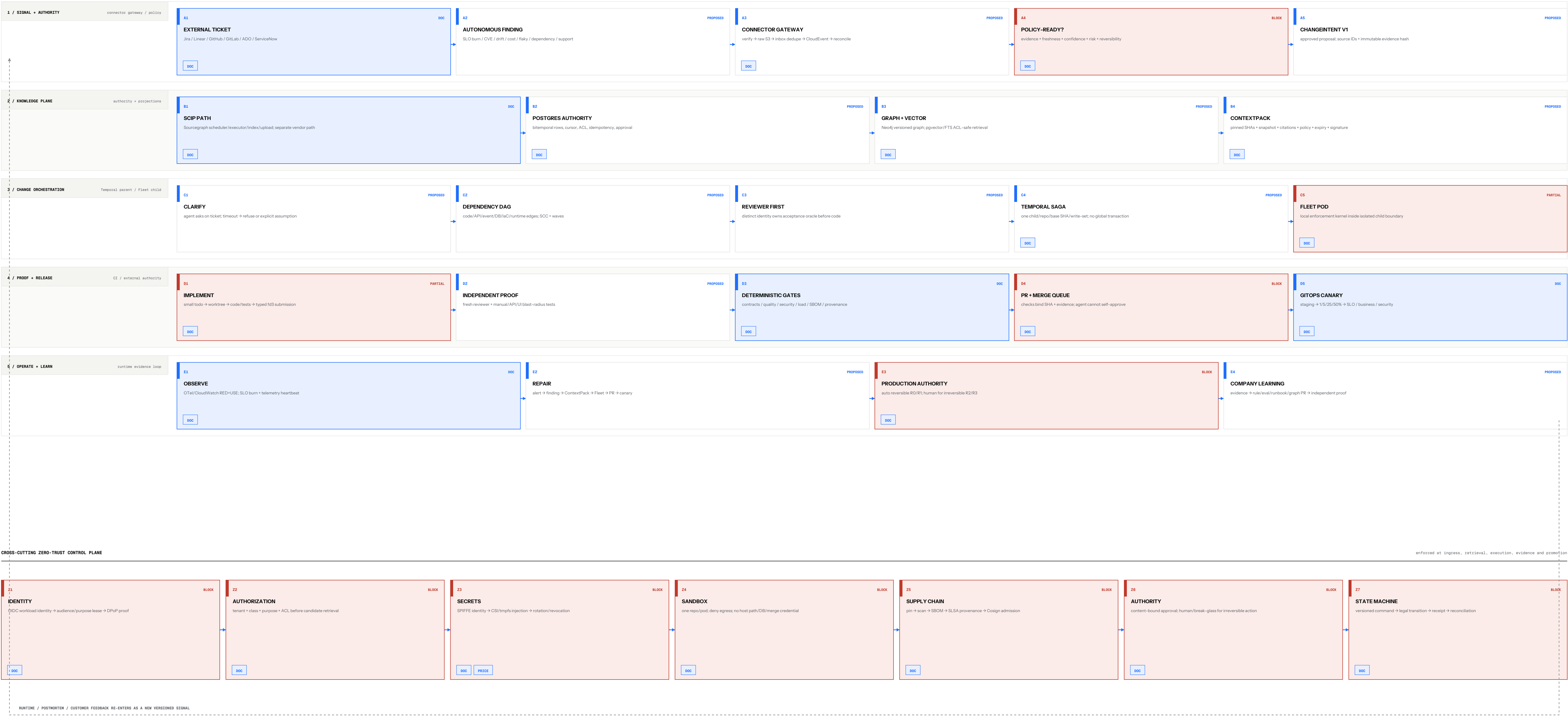
CONTROL / PROPOSED

EVIDENCE / LIVE

BLOCK / HUMAN

DATA / REPLAY

DOC = OFFICIAL SOURCE / PROPOSED = TARGET / LIVE = CHECKOUT EVIDENCE



Signals, ingestion and autonomous discovery

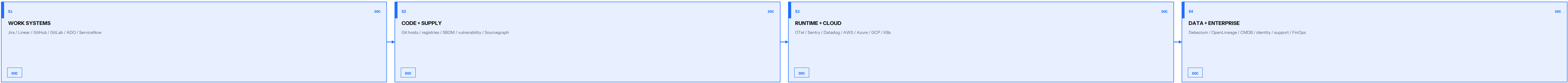
Webhooks reduce latency; cursor polls and snapshots establish completeness; agents only propose.

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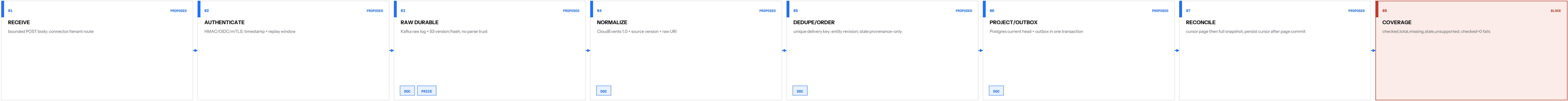
A / SOURCE-NEUTRAL CONNECTOR CONTRACT

webhook + cursor + snapshot + command/willteback



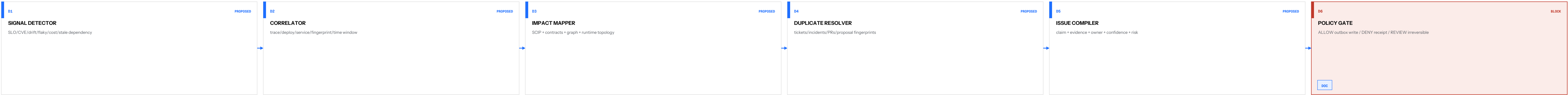
B / HOT PATH

ACK only after durable raw acceptance; enrichment stays asynchronous



C / AUTONOMOUS FINDING-TO-CHANGE LOOP

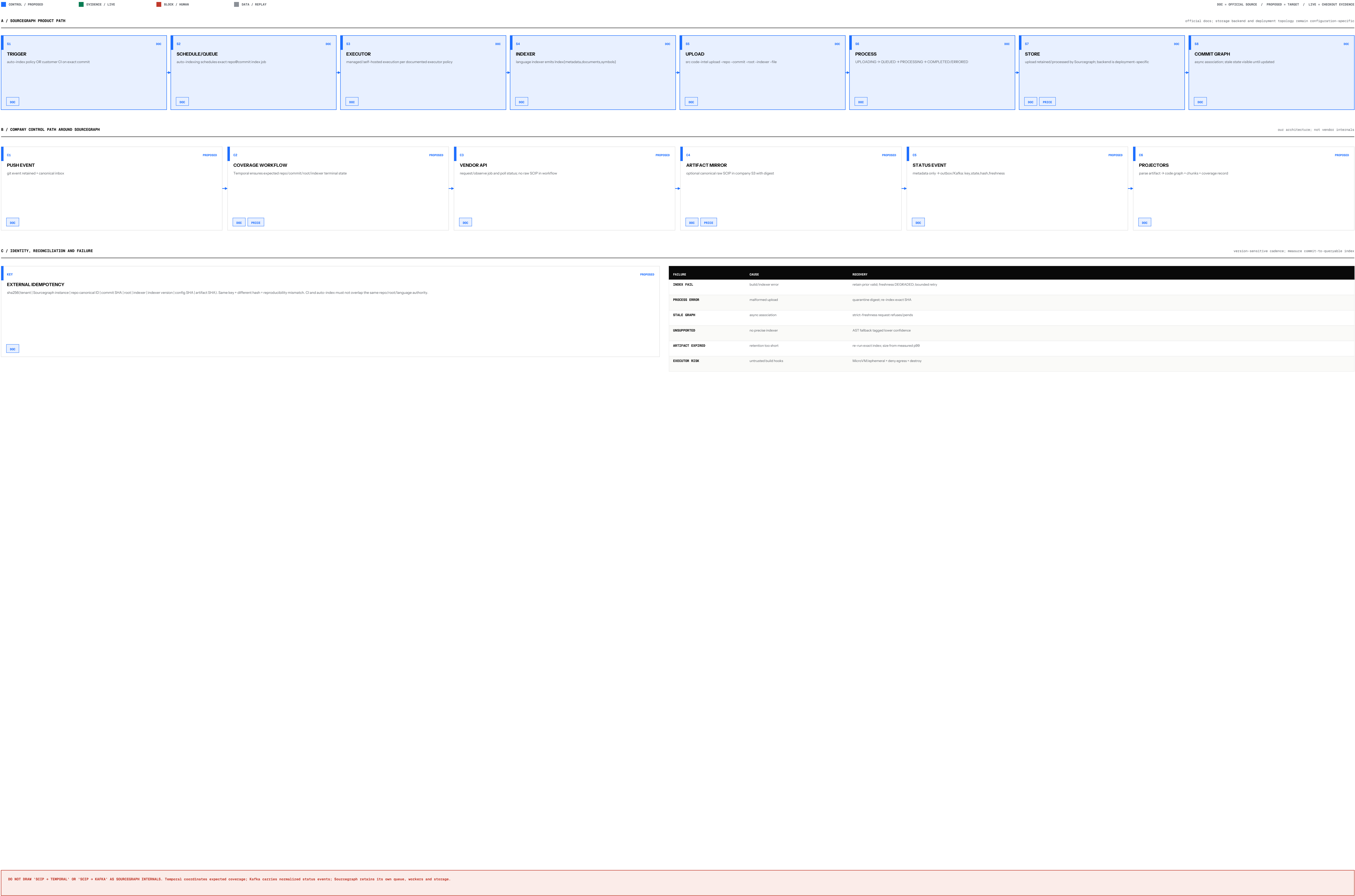
1read-only specialists behind deterministic state transitions



FATLURE	CAUSE	RECOVERY / PROOF
DUPLICATE	provider retry	unique inbox key; 2xx/no second effect
OUT OF ORDER	parallel delivery	source version compare; reconcile head
MISSING DELETE	no delete event	two complete scans; permission loss 1x delete
RATE LIMIT	429/quota	Retry ~After + tenant token bucket; cursor retained
SCHEMA DRIFT	payload changed	raw retained; quarantine; adapter version
WEBHOOK GAP	outage/expired hook	cursor + snapshot + coverage denominator

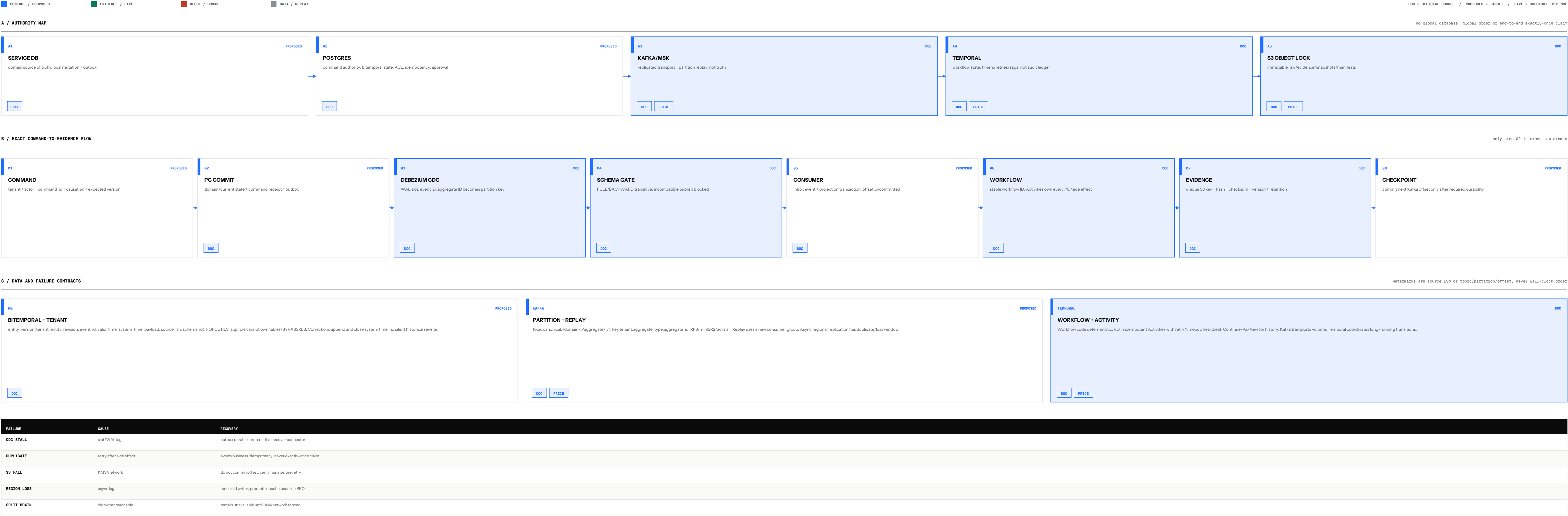
Sourcegraph SCIP: exact provenance and corrected arrows

SCIP does not go directly to Temporal or Kafka. Documented product behavior and company controls stay separate.



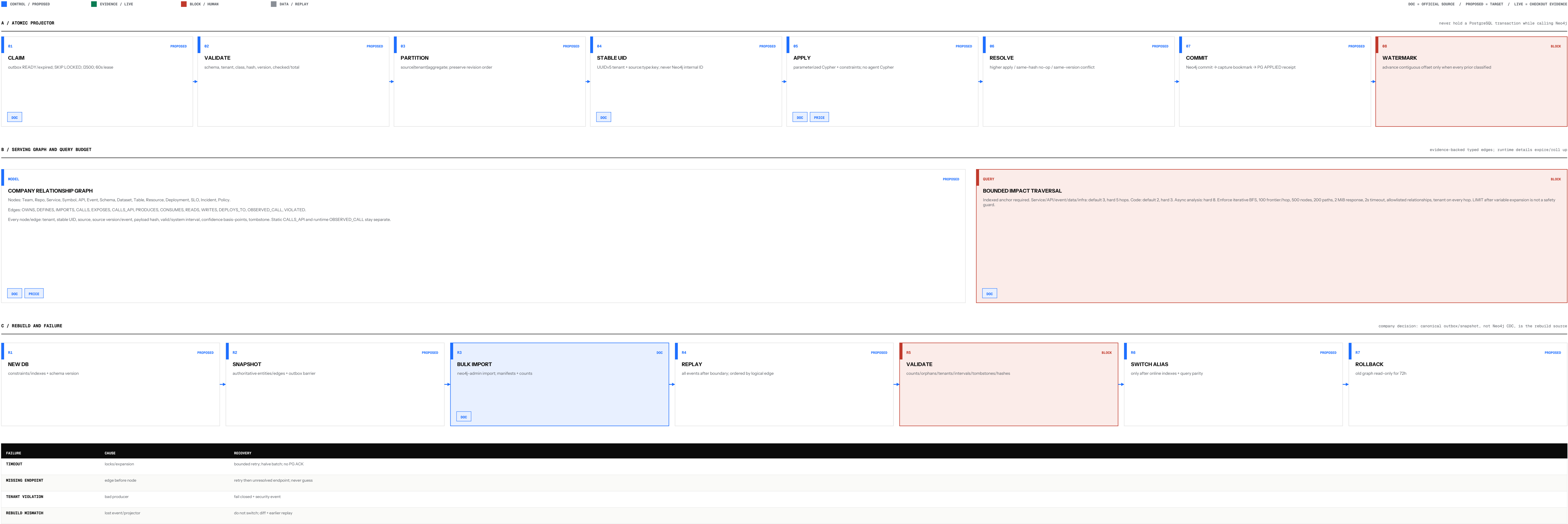
Canonical authority: Postgres, Kafka, Temporal and S3

Each system owns one concern. Only the PostgreSQL command transaction authorizes lifecycle transitions.



Neo4j projection: identities, ordering, conflicts and rebuild

Proposed company projector: Postgres resolves truth; Neo4j is treated as an idempotent disposable serving projection.



Chunk, vector, graph and signed ContextPack

Worked code example: structural atoms, ACL-before-candidate retrieval, hybrid rank, graph expansion and proof.

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A / INGESTION-TO-CONTEXTPACK

1824 dimensions default; 512 only after a protected eval win

01

PROPOSED

PIN
repo@SHA / doc ETag / contract digest / incident revision

02

DOC

PARSE
SCP definitions/occurrences + AST exact ranges

DOC

03

PROPOSED

CHUNK
exported/hot symbol 350-700 tokens; hard 1,200

04

BLOCK

SECURE
classification, ACL, secret/PII/injection; quarantine

05

PROPOSED

IDENTITY
logical UUIDv5 + immutable chunk version/content hash

06

PROPOSED

EMBED/INDEX
voyage-code=4/voyage-4 @1024 + PG FTS + HNSW

DOC

PRICE

07

BLOCK

RETRIEVE
authorize → exact anchors + FTS100 + vector100

DOC

08

PROPOSED

FUSE/EXPAND
RRF k=60 → typed graph expansion → rerank top40

DOC

09

PROPOSED

SELECT
MMR + topology coverage within 1024k tokens

10

PROPOSED

SIGN
ContextPack SHAa/napshots/citations/coverage/expiry

DOC

B / WORKED EXAMPLE

Fleet::graph::check_freshness from graph.rs

CHUNK

CANONICAL CODE ATOM
logical_chunk_id: UUIDv5(...)
chunk_version_id: sha256(...)
repo: fleet-rs@ commit SHA
path: keel/fleet/src/graph.rs
lines: 1025-1071
symbol: fleet:graph:check_freshness
signature + docs + complete function
token_count: 314
acl_policy_id / class / trust
content_hash / parser + version
edges: impact_command CALLS: CALLS tree_digest

PROPOSED

VECTOR
EMBEDDING + INDEXES

model: voyage-code=4
revision: pinned
dimensions: 1024
metric: cosine
vector: [-0.012, 0.002, ...]

Heaps: vector(1024): HNSW expression index may use halfvec(1024); rerank against full vector. PG FTS QIN weights exact symbol/path/signature/body. ACL relation is materialized before lexical. ANN, graph, cache or reranker receives candidates.

DOC

PRICE

PROPOSED

PACK
SIGNED RESULT MANIFEST

pack_id / change_set / child
actor + authorization digest
repo_sha_map + graph/foreknowledge snapshot
projector watermarks
lexical_k=100 / vector_k=100
rft_k=60 / rerank_k=40
token_budget=24000
coverage[checked_total,unknown,stable]
item[[]source URI, range, hash, ranks, graph paths]
truncation/threshold digest
issued/expires + manifest digest + Ed25519 signature

DOC

C / ACL, FRESHNESS AND EVALUATION

client-side post-filtering is never authorization

ACL

AUTHORIZE BEFORE CANDIDATE GENERATION

Validate CDC subject, tenant, purpose, clearance, groups. Materialize authorized source relation. Fine ACL sets use exact vector search; broad same-visibility partitions may use HNSW iterative scans. Every graph hop rechecks tenant/ACL. Empty allowlist, stale permission, dimension mismatch, missing citation or checked=0 refuses.

DOC

METRIC	GATE	PROOF
Recall13@100	≈0.95	internal 500 code + 500 company queries
nDOC@10	≈0.75	no source class regression >0.03
Exact MRR@10	≈0.85	symbol/API/incident targets
Topology coverage	≈0.90	definition/contract/caller/text/owner/incident
ACL leakage	0 / 10k pairs	tenant, revoked, poisoned, deleted
ContextPack p95	<2s cached	>8s cold; publish checked/total

Clarification, multi-repo impact and release DAG

Thirty repos and 100+ services require SCCs, compatibility waves and compensation—not a distributed transaction.

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A / PLAN BEFORE CODE

reviewer and acceptance oracle are assigned before implementation

P1

CONSUME

ticket/ChangeIntent exact version + evidence hash

PROPOSED

P2

CLARIFY

missing AC/NFR/owner/data/rollback → ticket questions

PROPOSED

P3

CONTEXT

signed per-child pack: sources, graph, contracts, incidents

PROPOSED

P4

IMPACT GRAPH

code/API/event/DB/taC/runtime edges + provenance

PROPOSED

P5

SCC COLLAPSE

cycles become one coupled release unit; topological DAG

PROPOSED

P6

COMPATIBILITY

version/adaptor or expand-migrate-contract

PROPOSED

P7

REVIEWER FIRST

distinct identity/model writes oracle + abuse/manual plan

PROPOSED

P8

ATOMIZE

smallest todo with repo/SHA/write-set/deps/tests/rollback

PROPOSED

P9

DISPATCH

Temporal parent starts dependency-ready Fleet children

PROPOSED

DOO

PRICE

B / WORKED MULTI-REPO WAVE

timeout field added to POST /charge

WAVE	REPO/UNIT	SURFACE	ACTION + PROOF	DEFENDS
W0	contracts	OpenAPI/event/Pact	add optional field; compatibility gate	none
W1	ledger-migrations	database	expand nullable column; lock proof	W0
W1	generated-sdk	client	regenerate compatible client	W0
W2	ledger-service	provider	read old/new; dual write; provider verify	W0,W1
W2	payments-api	consumer	new timeout fallback; client/unit/load tests	W0,W1
W3	environment	GitOps	feature flag + staged canary	all W2
W4	cleanup	all repos	contract old field only after adoption proof	outcome window

C / CHILD CONTRACT AND FAILURE

overlapping write-sets serialize; every changed base SHA invalidates context/approval

T000

ATOMIC CHILD SPECIFICATION

todo_id / parent / child / attempt; repo + immutable base SHA; branch; read/write sets; predecessor IDs; ContextPack digest + lease jti; acceptance predicates + exact test commands; risk + budget + max attempts; rollback/forward-fix; reviewer/oracle digest; expected evidence count. Parent success requires checked == total == required children.

PROPOSED

FAILURE	CAUSE	RECOVERY
MISSING CONTEXT	no owner/contract/class	clarify then refuse or exploit bounded assumption
GRAPH CYCLE	mutual dependency	collapse SCC; one coupled wave
BREAKING API	consumer incompatible	version/adaptor; Pact can-i-deploy
DESTRUCTIVE DB	drop/lock risk	expand-migrate-contract; human boundary
PARTIAL MERGE	child fails	pause saga; preserve success; compensate/forward fix

Fleet inside the enterprise SDLC

Fleet is the per-repo enforcement kernel. Remote context, Kubernetes isolation and multi-repo fan-in are target extensions.

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A / ONE CHILD EXECUTION

F1

ENVELOPE

tenant,parent,repo SHA,write-set,risk,idempotency

PROPOSED

DOC

F2

CONTEXT LEASE

OIDC → audience=bound 10-15m OAuth token; no refresh

PROPOSED

DOC

F3

CHILD JOB

one repo, worktrees/*child*, pinned image; deny egress

PROPOSED

DOC

F4

FLEET GATE

clean tree + accepted SOW + scoped role/model

LIVE

F5

WORKER

Codex/Claude adapter; stdin closed; no ledger/state/socket env

PARTIAL

DOC

F6

FD3 SUBMIT

typed note/done/refuse; child cannot stamp authority facts

LIVE

F7

FREEZE

diff + BLAKE3 artifact; empty diff invariant refusal

LIVE

F8

VERIFY

distinct identity; blind suite + adequacy + blast radius + rollback

PARTIAL

F9

RECEIPT

local stamps live; enterprise evidence schema not yet enforced

PARTIAL

F10

EXTERNAL PR

controller opens PR/checks; host owns merge authority

PROPOSED

DOC

PRICE

separate company Context MCP and local Fleet studio MCP trust planes

B / CONTRACTS AND ISOLATION

PACK

TICKETENVELOPE + CONTEXTPACK

Ticket version, actor, purpose, risk, repos/base SHAs, dependency edges, integer token/cost/time budgets, accepted SOW and idempotency key. Per-child signed ContextPack pins graph/knowledge snapshots, source digests, policy, repo SHA map, coverage and expiry. Retrieval is remote read-only MCP; local file tools remain stdio.

PROPOSED

DOC

PRICE

P00

ONE JOB / ONE REPO / ONE ATTEMPT

restartPolicy Never; backoffLimit 0; automount service token false; non-root; read-only root; seccomp RuntimeDefault; drop ALL; resource/PID/deadline limits; size-limited emptyDir; default/-deny network. Trusted sidecar brokers lease and uploads evidence. No host path, Docker socket or shared writable Git store.

PROPOSED

DOC

RECEIPT

WORKER VS PARENT AUTHORITY

LIVE LOCAL: t45-only [schema, version, kind, body]; no ledger/state/socket path; parent stamps seq/hash/time/actor/model/exit/artifact; refusal receipt precedes exit; immutable acceptance tests. Typed exits: 0 ok, 3 environment, 6 invariant, 7 refusal, 8 mismatch. TARGET SCHEMA: parent/child IDs, SHAs, ContextPack/lease hashes, checked/total and cost source; not yet contract-enforced.

PARTIAL

agents never receive database, graph, vector, ledger or merge credentials

C / HONEST STATUS

current checkout evidence is not a production-cloud proof

CURRENT FLEET	STATUS	ENTERPRISE EXTENSION	STATUS
studio MCP + scoped files	LIVE	remote HTTP Context MCP + OAuth	PROPOSED
local worktree + fd3	LIVE	ephemeral Kubernetes child	PROPOSED
Codex/Claude subprocess	PARTIAL	real-adapter corpus + deterministic config	UNPROVEN
SQLite tree-sitter graph	LIVE	cross-repo graph/vector/runtime context	PROPOSED
hash-linked local receipts	LIVE	signed WORM checkpoints + HA	PROPOSED
swarm lifecycle	PARTIAL	30-repo parent/fan-in + PR authority	PROPOSED

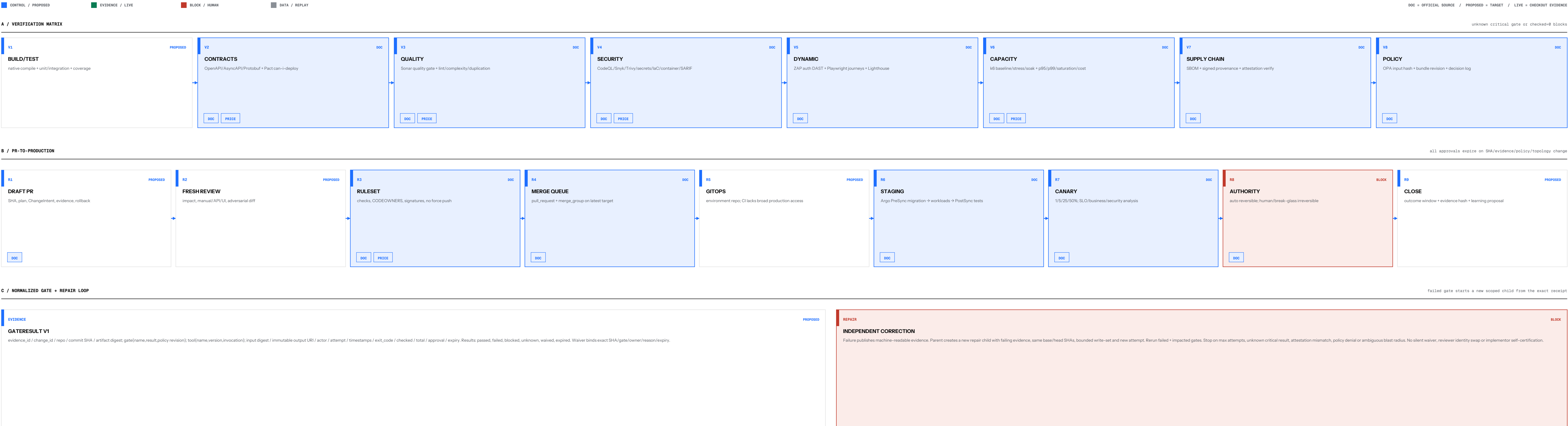
PROOF BOUNDARY: current targeted bash tests/acceptance/p0.sh = 34 passed / 0 failed on 25 Aug 2026. This is not the full verify.sh exit. P1-P4, remote MCP, real model-adapter corpus and distributed operation remain unearned. Fleet stays local/partial where evidence stops.

DOC/PRICE badges and linked table cells open official sources.

DEVX CONFIDENTIAL / PAGE 08

Independent review, CI gates, merge and release

Model prose is not evidence. Each gate binds inputs, version, invocation, exit code, denominator and immutable output.



SLO, capacity, incident repair, DR and company learning

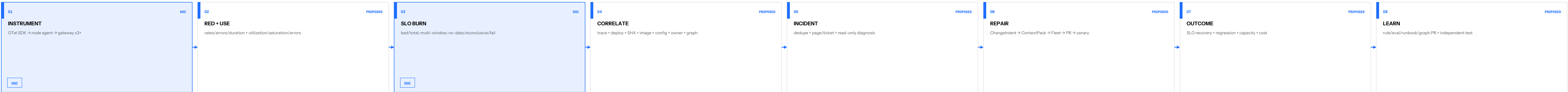
Runtime hooks create findings; the same SDLC loop repairs production through protected PRs and canaries.

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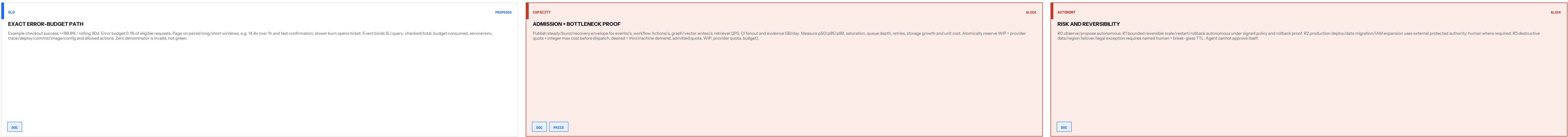
A / RUNTIME-TO-REPAIR LOOP

one SLO authority; telemetry loss pauses promotion



B / SLO, CAPACITY AND AUTONOMY

scale from measured bottleneck; never autoscale past authority or cost reservations



C / DISASTER RECOVERY

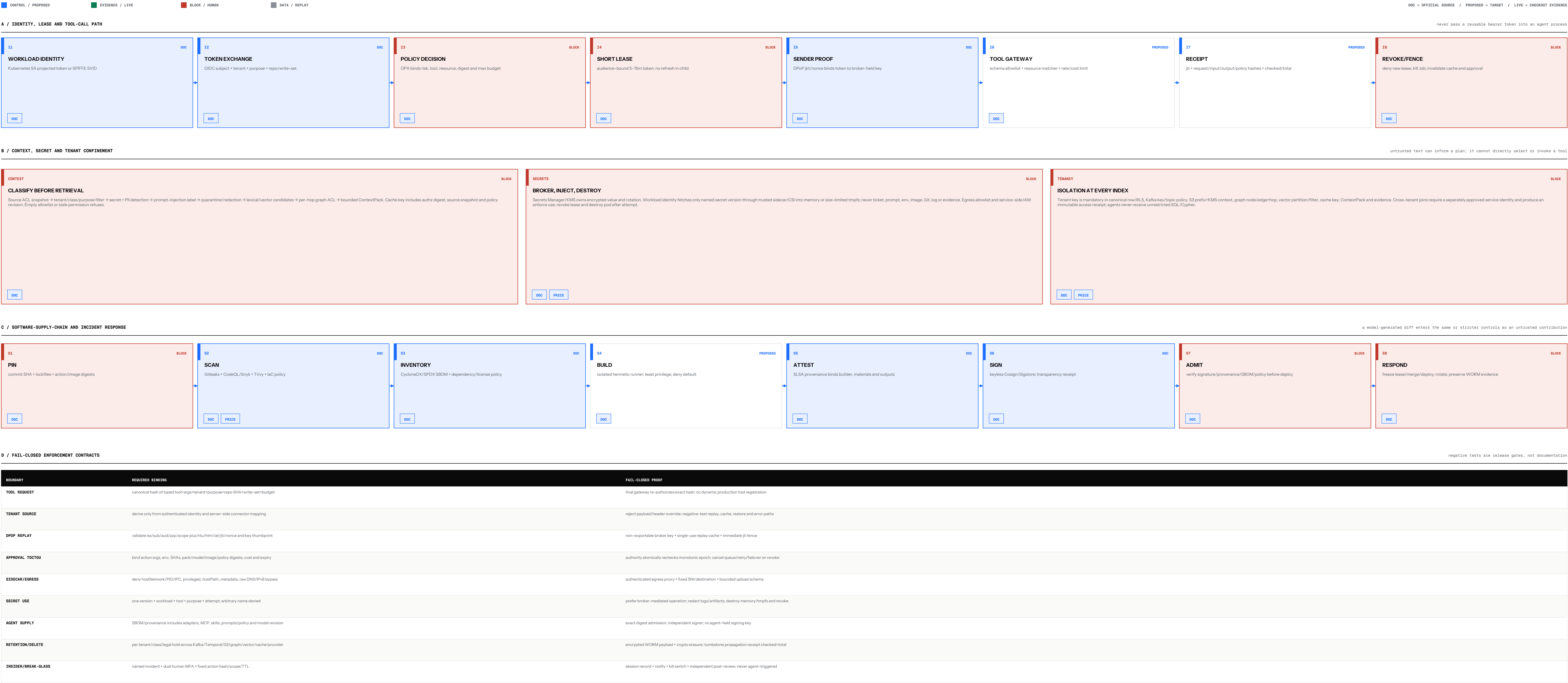
recovery is a tested workflow with checked/total, not a backup-success badge

PLANE	RECOVERY MECHANISM	ACCEPTANCE PROOF
POSTGRES	P11R + cross-region replica	new DB known LSN/time; validate CDC slot continuity
KAFKA	R33/AZ + async replication	fence old region; RPO=lag; dedupe/reconcile
TEMPORAL	multi-region/custom or active-passive	region token fences Activity side effects
MED4J	snapshot + outbox replay	new DB counts/hashes/query parity; alias switch
VECTOR	rebuild from chunks/model revision	ACL/recall/stale-vector eval before cutover
S3	Object Lock + cross-account CBR	restore/checksum/manifest-chain drill
FLEET	durable parent + fresh child	receipt reconciliation; never reuse dirty workspace

COMPANY LEARNING IS NOT CHAT MEMORY: immutable incident/change evidence → verified lesson candidate → owner + scope + expiry → policy/eval/runbook/gzaph PR → independent test → versioned publication → retrieval only after ACL and freshness checks.

Security and trust boundaries across every agent hop

Identity is workload-bound; context is pre-authorized; tools are deterministic; irreversible authority stays external.



16

TOOL GATEWAY

schema allowlist + resource matcher + rate/cost limit

DOC

17

RECEIPT

jti + request/input/output/policy hashes + checked/total

PROPOSED

18

REVOKE/FENCE

deny new lease; kill job; invalidate cache and approval

DOC

B / CONTEXT, SECRET AND TENANT CONFINEMENT

untrusted text can inform a plan; it cannot directly select or invoke a tool

CONTEXT

CLASSIFY BEFORE RETRIEVAL

Source ACL snapshot → tenant/class/purpose filter → secret + PII detection → prompt-injection label → quarantine/redaction → lexical/vector candidates → per-hop graph ACL → bounded ContextPack. Cache key includes authz digest, source snapshot and policy revision. Empty allowlist or stale permission refusals.

DOC

SECRETS

BROKER, INJECT, DESTROY

Secrets Manager/KMS owns encrypted value and rotation. Workload identity fetches only named secret version through trusted sidecar/CSI into memory or size-limited tmpfs; never ticket, prompt, env, image, Git, log or evidence. Egress allowlist and service-side IAM enforce use; revoke lease and destroy pod after attempt.

DOCPRICE

TENANCY

ISOLATION AT EVERY INDEX

Tenant key is mandatory in canonical row/RLS, Kafka key/topic policy, S3 prefix+KMS context, graph node/edge+hop, vector partition/filter, cache key, ContextPack and evidence. Cross-tenant joins require a separately approved service identity and produce an immutable access receipt; agents never receive unrestricted SQL/Cypher.

DOCPRICE

C / SOFTWARE-SUPPLY-CHAIN AND INCIDENT RESPONSE

a model-generated diff enters the same or stricter controls as an untrusted contribution

81

PIN

commit SHA + lockfiles + action/image digests

DOC

82

SCAN

GitReeks + CodeQL/Snyk + Trivy + IaC policy

DOCPRICE

83

INVENTORY

CycloneDX/SPDX SBOM + dependency/license policy

DOC

84

BUILD

isolated hermetic runner; least privilege; deny default

DOC

85

ATTEST

SLSA provenance binds builder, materials and outputs

DOC

86

SIGN

keyless Cosign/Signstore; transparency receipt

DOC

87

ADMIT

verify signature/provenance/SBOM/policy before deploy

DOC

88

RESPOND

freeze lease/merge/deploy; rotate; preserve WORM evidence

DOC

D / FAIL-CLOSED ENFORCEMENT CONTRACTS

negative tests are release gates, not documentation

BOUNDARY	REQUIRED BINDING	FAIL-CLOSED PROOF
TOOL REQUEST	canonical hash of typed tool+args+tenant+purpose+repo SHA+write-set+budget	final gateway re-authorizes exact hash; no dynamic production tool registration
TENANT SOURCE	derive only from authenticated identity and server-side connector mapping	reject payload/header override; negative-test replay; cache, restore and error paths
DPoP REPLAY	validate ts/sub/aud/azp/scope plus htt/htm/iat/jti/nonce and key thumbprint	non-exportable broker key + single-use replay cache + immediate jti fence
APPROVAL TOCTOU	bind action args, env, SHAAs, pack/model/image/policy digests, cost and expiry	authority atomically rechecks monotonic epoch; cancel queue/retry/failover on revoke
SIDECAR/EGRESS	deny hostNetwork/PID/IPC, privileged, hostPath, metadata, raw DNS/IPv6 bypass	authenticated egress proxy + fixed SNIP/destination + bounded upload schema
SECRET USE	one version + workload + tool + purpose + attempt; arbitrary name denied	prefer broker-mediated operation; redact logs/artifacts; destroy memory/tmpfs and revoke
AGENT SUPPLY	SBOM/provenance includes adapters, MCP, skills, prompts/policy and model revision	exact digest admission; independent signer; no agent-held signing key
RETENTION/DELETE	per tenant/class/legal hold across Kafka/Temporal/S3/graph/vector/cache/provider	encrypted WORM payload + crypto erasure; tombstone propagation receipt checked-total
INSIDER/BREAK-GLASS	named incident + dual human MFA + fixed action hash/scope/TTL	session record + notify + kill switch + independent post-review; never agent-triggered

FORBIDDEN: prompt-selected raw shell/SQL/Cypher, bearer passthrough, child refresh token, secret in model context, ACL-after-retrieval, mutable image/action, self-approval, unlogged override, or destructive autonomous action. Break-glass is named, MFA-protected, time-boned, scope-bound and post-reviewed.

Executable lifecycle state machine and canonical contracts

Cards become implementation only when each transition has one command, one guard, one authority and one receipt.

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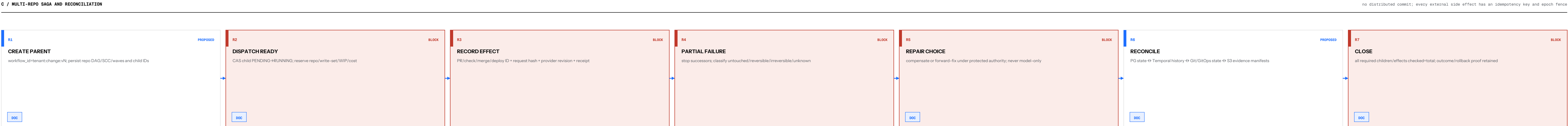
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A / LEGAL TRANSITIONS			
TRANSITION	COMMAND + AUTHORITY	ATOMIC GUARD + EVIDENCE	TIMEOUT / RECOVERY
SIGNAL_ACCEPTED → REFINED	SignalNormalized / connector	raw URI+hash; source revision; inbox idempotency; tenant from identity	retry adapter; quarantine schema drift
REFINED → CONTEXT_READY	ContextPackedIssued / context gateway	ACL+freshness; graph/vector watermark; checked-total; signed expiry	refresh or refuse; never state fallback
REFINED → NEEDS_CLARIFICATION	ClarificationRequested / planner	missing AC/NFR/owner/data/rollback recorded as typed questions	ticket comment; timer; no worker dispatch
NEEDS_CLARIFICATION → REFINED	AnswerRecorded / connector	expected ticket revision; answer author; assumption policy decision	conflict creates new signal version
CONTEXT_READY → PLANNED	PlanCompleted / planner	pinned graph snapshot; unknown edges; SCCs; waves; child read/write sets	unknown critical edge blocks
PLANNED → ORACLE_LOCKED	ReviewOracleStored / reviewer authority	distinct identity; acceptance/manual/abuse plan digest; no implementation diff	expiry on plan/context/policy change
ORACLE_LOCKED → IMPLEMENTING	ChildDispatched / Temporal parent	base SHA+pack+oracle+lease current; WIP/cost reserved; repo lock	idempotent child ID; bounded retry
IMPLEMENTING → VERIFYING	SubmissionFrozen / Fleet parent	fdS typed done; nonempty diff; artifact+receipt hash; exact exit taxonomy	refusal/failure receipt before terminal
VERIFYING → PR_READY	GateSetPassed / proof controller	independent review; required evidence checked-total; no unknown critical gate	new repair child; max attempts
PR_READY → MERGE_QUEUED	MergeRequested / protected Git authority	head SHA/evidence/oracle/policy/topology unchanged; approvals current	rebase invalidates and returns VERIFYING
MERGE_QUEUED → STAGING	MergedAndPromoted / GitOps authority	merge-group SHA; provenance/SBOM/signature; environment commit	queue failure returns PR_READY
STAGING → CANARY → RELEASED	PromotionStep / release authority	migration proof; 1/5/25/50%; SLO+business+security windows	auto rollback only if signed reversible action
RELEASED → OBSERVING → CLOSED	OutcomeAccepted / outcome controller	outcome window; SLO/capacity/cost; learning proposal; evidence hash	regression opens versioned repair signal
ANY → BLOCKED	PolicyDenied/Ambiguous/BudgetExceeded / controller	reason code + missing evidence + checked-total + owner + expiry	no silent success; explicit resume command
SIDE_EFFECTED → COMPENSATING	SagaRepairRequested / protected authority	completed effects classified; compensation/forward-fix owner and proof	fence epoch; never infer rollback from timeout

B / VERSIONED CORE CONTRACTS

canonical JSON, sorted keys, integer budgets, hashes over bytes; schema registry blocks incompatibility

COMMAND COMMANDENVELOPE V1 command_id / tenant / change_id / expected_state_version / type / actor identity / purpose / risk / aggregate ID / exact resource+args hash / idempotency key / causation+correlation IDs / policy approval digest / max integer cost+time / issued+expires / signature. PostgreSQL compares expected version and appends state+outcome receipt atomically. DOC	SIGNAL SIGNAL + FINDING + CHANGEINTENT V1 Signal: source instance/object/revision, raw URI/hash, occurred/received, delivery key, connector/schema version, delete/permission state. Finding: detector/version, subject, evidence refs, reproduction, confidence basis-points, dedupe fingerprint, checked/total, TTL. Changeintent: accepted facts/assumptions, AC/NFR, repos, risks, allowed actions and policy receipt. DOC	WRITEBACK PROVIDERCOMMAND + TRANSITIONRECEIPT V1 Writeback: provider, object, expected revision, operation(comment,status,label,link), idempotency key, retry budget; conflict re-ingests latest version. Receipt: before/after state versions, command/event IDs, actor authority, input/output/evidence hashes, policy approval digests, side-effect key, exit code, checked/total, attempt, epoch and expiry. DOC
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INVALIDATION: ticket revision, base/head SHA, graph or ACL watermark, ContextPack digest/expiry, oracle, policy bundle, model/image/tool digest, approval epoch or topology change invalidates every downstream lease and approval. Reconciliation classifies UNKNOWN as blocked, never as success.

Tools, official links and public pricing snapshot

Checked 25 Aug 2026. Public list prices are not enterprise TCO; custom and region-sensitive items stay explicit.

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TOOL	PRICING UNIT	PUBLIC LIST USD	AS OF	BOUNDARY
Sourcegraph Enterprise	platform/contract	Starts \$19K	2026-08-25	public starting price; term/support custom
Temporal Cloud	Actions + storage	\$50/1M Actions + \$100/mo AWS path	2026-08-25	AWS path separate; storage/multi-region confirm
Neo4j Aura BC	GB memory/month	\$146/GB/mo; 2GB minimum	2026-08-25	VDC/CMEK/private endpoints custom
Postgres + pgvector	license	\$0 software license	2026-08-25	HA compute/storage/ops extra
MSK	broker/GB/network	region and tier usage based	2026-08-25	calculator + support/egress
Confluent Cloud	eCKU/GB/network	region and tier usage based	2026-08-25	calculator + contract
S3 Object Lock	GB-months/requests/CRR	region/class usage based	2026-08-25	retention/KMS/egress sized
GitHub Enterprise	seat/month	\$21/user/mo first 12 months	2026-08-25	EMU/residency/support/actions
GitHub security	active committer/month	\$30 Code; \$18 Secret Protection	2026-08-25	90-day unique committer
Sonatype Cloud Team	private LOC/month	Starts \$34/mo up to 100K LOC	2026-08-25	Enterprise/Server/DC custom
Snyk Team / Ignite	contributing developer	\$25/mo; \$1,260/year	2026-08-25	Enterprise/Broker custom
PactFlow Team	integration plan/month	\$127 monthly; \$115.42 annual	2026-08-25	Enterprise/on-prem custom
Craftera Cloud K8 Pro	VUH + platform	\$0.15/VUH + \$19/mo	2026-08-25	enterprise/EYOC custom
HCP Terraform Essentials	resource-hour	verify current public rate	2026-08-25	linked source lacks stable quoted rate
Voyage embeddings	1M tokens after free tier	code-4 \$0.12; voyage-4 \$0.06	2026-08-25	first 200M/account free; revisions vary
ChatGPT Business	user/month	\$20 annual; \$25 monthly	2026-08-25	2-seat minimum; Codex access included
Codex/API usage	model tokens/credits	usage based; model-specific	2026-08-25	workspace credits/API billing separate
Anthropic API	1M input/output tokens	Opus5 \$5/\$25; Sonnet5 \$2/\$10; Haiku4.5 \$1/\$5	2026-08-25	cache/region/agent modifiers
AWS Secrets Manager	secret-month + API calls	\$0.40/secret-mo + \$0.05/10K calls	2026-08-25	rotation/replication/region usage
AWS KMS	key-month + requests	\$1/customer key-mo; requests usage-based	2026-08-25	rotation/region/HSM/sign usage
Core OSS control stack	license	\$0 OSS license	2026-08-25	compute/storage/support/operations extra

B / COST MODEL AND BUY/BUILD

do not publish a fake single TCO

FORMULA	BOUNDARY	PROPOSED
MONTHLY COST INPUTS Seats + Sourcegraph contract + workflow Actions/storage + event compute/storage/network + Postgres + graph GB + object storage/requests/replication + CI runners + security committers/LOC + test VUH + embedding/rerank/model tokens/tools + support + 24x7 operations. Size from engineers, LOC, events/day, nodes/edges, chunks, OPS, agent runs/tokens, CI fanout, evidence GB/day, retention, regions and RPO/RTQ.	RECOMMENDED GREENFIELD CHOICE BUY managed data/workflow/code/security/identity/KMS/Kubernetes primitives. BUILD connector SDK, canonical contracts, bitemporal projection, graph projector, Context Gateway/Pack, dependency planner, Fleet cloud parent/child, evidence ledger, policy state machine and eval/learning pipeline. DO NOT BUILD broker, graph DB, CI engine, analyzer, LSP, signer, scheduler or global transaction coordinator.	

Individual official-source catalog

Every named implementation product has its own clickable primary documentation or specification link.

CONTROL / PROPOSED EVIDENCE / LIVE BLOCK / HUMAN DATA / REPLAY

DOC = OFFICIAL SOURCE / PROPOSED = TARGET / LIVE = CHECKOUT EVIDENCE

TOOL / SPEC	ROLE IN ARCHITECTURE	SOURCE
Jira webhooks	work intake	OFFICIAL
Linear webhooks	work intake	OFFICIAL
GitHub webhooks	code/work intake	OFFICIAL
GitLab webhooks	code/work intake	OFFICIAL
Azure DevOps service hooks	code/work intake	OFFICIAL
Bitbucket webhooks	code/work intake	OFFICIAL
ServiceNow Table API	enterprise work intake	OFFICIAL
CloudEvents	canonical event envelope	OFFICIAL
Debezium	CDC	OFFICIAL
Debezium outbox	transactional outbox	OFFICIAL
OpenLineage	data lineage	OFFICIAL
Sourcgraph auto-index	precise code index	OFFICIAL
Sourcgraph executors	index execution	OFFICIAL
SCIP schema	code index artifact	OFFICIAL
PostgreSQL RLS	canonical authorization	OFFICIAL
PostgreSQL FTS	lexical retrieval	OFFICIAL
pgvector	vector retrieval	OFFICIAL
Kafka	ordered partition transport	OFFICIAL
Confluent Schema Registry	schema compatibility	OFFICIAL
Temporal	durable orchestration	OFFICIAL
S3 Object Lock	immutable evidence	OFFICIAL
Neo4j	erving graph	OFFICIAL
Voyage AI	code/prose embeddings	OFFICIAL
MCP specification	agent context/tools	OFFICIAL
MCP authorization	resource-server auth	OFFICIAL
OpenID Connect	workload federation	OFFICIAL
DPoP RFC 9449	sender-constrained token	OFFICIAL
SPFFE	workload identity	OFFICIAL
Secrets Store CSI	secret delivery	OFFICIAL
AWS Secrets Manager	secret authority	OFFICIAL
AWS KMS	key authority	OFFICIAL
Kubernetes Jobs	ephemeral child	OFFICIAL
Kubernetes NetworkPolicy	deny-default-egress	OFFICIAL

TOOL / SPEC	ROLE IN ARCHITECTURE	SOURCE
Git worktree	isolated checkout	OFFICIAL
OpenAI Codex	implementation adapter	OFFICIAL
Claude Code	implementation adapter	OFFICIAL
GitHub pull requests	PR authority	OFFICIAL
GitHub rulesets	protected policy	OFFICIAL
GitHub merge queue	latest-target validation	OFFICIAL
Pact	consumer contract proof	OFFICIAL
SonarQube	quality gate	OFFICIAL
CodeQL	code scanning	OFFICIAL
Snyk	dependency/code security	OFFICIAL
Trivy	repo/IaC/container scan	OFFICIAL
GitLeaks	secret scan	OFFICIAL
OWASP ZAP	authenticated DAST	OFFICIAL
Playwright	journey/manual proof	OFFICIAL
Lighthouse CI	web performance	OFFICIAL
Grafana K6	capacity tests	OFFICIAL
CycloneDX	SBOM	OFFICIAL
SLSA 1.2	provenance	OFFICIAL
Cosign/Signstore	artifact signing	OFFICIAL
GitHub attestations	build attestation	OFFICIAL
OPA	deterministic policy	OFFICIAL
OpenTelemetry	telemetry	OFFICIAL
AWS CloudWatch	runtime monitoring	OFFICIAL
Sentry	error monitoring	OFFICIAL
Datadog	observability	OFFICIAL
Argo CD	GitOps sync	OFFICIAL
Argo Rollouts	canary analysis	OFFICIAL
SOLite	local Fleet graph/state	OFFICIAL
tree-sitter	local parsing	OFFICIAL
HCP Terraform	IaC collaboration	OFFICIAL
Presidio	PII detection	OFFICIAL
Google SRE burn rates	SLO alert method	OFFICIAL

SOURCE RULE: product behavior uses vendor docs/specifications; company-owned topology, SLOs, limits and orchestration are labelled PROPOSED; list price is never presented as enterprise T00. Broken or redirected links fail the publication gate.

CTO acceptance: when this architecture becomes real

The proposal is rejected until every requirement has hash-pinned evidence and a published denominator.

CONTROL / PROPOSED

EVIDENCE / LIVE

BLOCK / HUMAN

DATA / REPLAY

DOC = OFFICIAL SOURCE / PROPOSED = TARGET / LIVE = CHECKOUT EVIDENCE

ID	REQUIREMENT	FALSIFIABLE TEST	REJECT WHEN
V1-01	Source completeness	manifest vs API inventory + replay + snapshot	any unexplained omission/permission gap
V1-02	Autonomous discovery	seed code/runtime/security defects across languages	precision/recall absent or non-reproducible
V1-03	SCIP correctness	pinned upload/query hashes + freshness state	unsupported vendor-internal arrow/claim
V1-04	Temporal	kill worker; duplicate start; retry; signal; replay	repeated side effect/unbounded retry/history
V1-05	Multi-repo	shared-contract change across all consumers	partial merge without saga/compensation
V1-06	Graph	rebuild raw; no orphan; source SHA/range per edge	mixed revisions/irreproducible projection
V1-07	Vector	dimension/metric, ACL, delete, exact recall	cross-tenant/stale/united result
V1-08	Security	policy bypass, secret exfiltration, OIDO broadening	privilege expansion/untraceable decision
V1-09	SLO/capacity	peak+burst ingress/workflow/graph/vector/CI	no p50/p95/p99/saturation/cost/backpressure
V1-10	DR	region loss, restore, replay, partial merge	RPO/RTO prose-only or duplicate action
V1-11	Authority	self-approval/direct push/stale approval reuse	forbidden transition succeeds
V1-12	Fleet	real commit/adapters/corpus/oracle/exit codes	local/placeholder marketed as cloud proof

8 / INDEPENDENT VERIFIERS

run after implementation and before autonomy expansion

ARCHBLOCK

ARCHITECTURE VERIFIER
Every arrow names control/data/evidence path, API/schema, owner, delivery semantics, retry, idempotency and recovery. Every projection names authority and rebuild source. Every multi-repo wave exposes partial success and compensation. Any ambiguous arrow rejects.

SECBLOCK

SECURITY/AUTONOMY VERIFIER
Reject ACL-after-retrieval, bearer passthrough, child refresh tokens, mutable images, self-approval, unrestricted SQL/Cypher/shell/network, missing tenant/class, stale approval, missing break-glass TTL or destructive autonomous action.

CONRBLOCK

EVIDENCE/PRICING VERIFIER
Reject secondary-source implementation claims, stale/unitless prices, list price represented as TCO, unlinked tools, vendor SLA shown as estate SLO, unsupported Fleet capability, or any green result without raw artifact + checked/total.

FIRST 90 DAYS: connector SDK + evidence contracts + Postgres/S3 authority + Sourcegraph + graph/vector projectors + Context Gateway/leases + Fleet Kubernetes child + GitHub PR/CI/GitOps + autonomous finding loop + capacity/DR + risk-tiered autonomy expansion.